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SCHOOL OF ENGINEERING AND MATERIALS SCIENCE

**EMS690U: Integrated Design Project**  
**Assessment 4**

**Individual Detailed Design Report**

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**Design and Evaluation of a Constraint-Based  
Emergency Landing Guidance System for a  
Heavy-Lift Cargo Unmanned Aerial Vehicle  
(UAV)**

|              |                    |
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## **Abstract**

Heavy-lift cargo UAVs operating beyond visual line of sight over populated terrain pose significant third-party risk if a mid-mission failure triggers an uncontrolled landing. Existing solutions span passive recovery hardware, perception-led site detection, and reactive trajectory re-planners, treating site selection, trajectory generation, and safety monitoring as loosely coupled modules. This report presents a constraint-based emergency landing guidance system for a hybrid tilt-rotor cargo UAV (500 kg payload, 200 km range, 2500 kg MTOW). The guidance layer formulates emergency landing as a constrained decision-making problem within a unified pipeline. A weighted-sum MCDM ranks candidate sites by reachability, ground risk, obstacle clearance, surface quality, wind alignment, and energy feasibility. A 3-DOF kinematic generator is combined with a runtime safety monitor that clamps descent rate, lateral error, and approach energy against bounds derived from a landing-gear impact-energy budget. Simulation of a cruise-stage propulsion failure over 200 Monte Carlo runs eliminates all safety-envelope violations (200/200 to 0/200) and reduces peak impact energy by 93.4% (341 kJ to 22.5 kJ), with a 94.2% reduction in mean peak energy per run, relative to an unsupervised waypoint baseline. Benchmarking against Cesetti et al. (2010) confirms improved applicability to failure-state operation. Limitations, regulatory implications, and future hardware-in-the-loop validation are discussed.

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## List of Symbols

| Symbol                 | Description                                                     | Units                                        |
|------------------------|-----------------------------------------------------------------|----------------------------------------------|
| $A_{\text{imp}}$       | Effective impact area on the ground                             | $\text{m}^2$                                 |
| $c$                    | Landing-gear damping coefficient                                | $\text{N}\cdot\text{s}/\text{m}$             |
| $c_i$                  | Raw value of the $i^{\text{th}}$ criterion for a candidate site | varies                                       |
| $\hat{c}_i$            | Min-max normalised value of the $i^{\text{th}}$ criterion       | [0, 1]                                       |
| $D$                    | Reachability cost (distance or energy to reach a site)          | $\text{m}$ or $\text{J}$                     |
| $E$                    | Kinetic impact energy at touchdown                              | $\text{J}$                                   |
| $E_{\text{available}}$ | Remaining on-board energy available for the manoeuvre           | $\text{J}$                                   |
| $E_{\text{max}}$       | Maximum permitted impact energy set by safety budget            | $\text{J}$                                   |
| $e_{\text{max}}$       | Maximum permitted lateral tracking error                        | $\text{m}$                                   |
| $e_{xy}$               | Horizontal cross-track tracking error                           | $\text{m}$                                   |
| $g$                    | Gravitational acceleration                                      | $\text{m}/\text{s}^2$                        |
| $h$                    | Altitude above ground level                                     | $\text{m}$                                   |
| $h_{\text{min}}$       | Minimum obstacle clearance margin                               | $\text{m}$                                   |
| $J$                    | Weighted multi-criteria site cost                               | dimensionless                                |
| $k$                    | Effective landing-gear stiffness                                | $\text{N}/\text{m}$ ( $\text{MN}/\text{m}$ ) |
| $k_z$                  | Vertical tracking gain                                          | $1/\text{s}$                                 |
| $L/D$                  | Lift-to-drag ratio (best glide)                                 | dimensionless                                |
| $m$                    | Vehicle mass (instantaneous)                                    | $\text{kg}$                                  |
| $N$                    | Landing-gear sizing load factor                                 | dimensionless                                |
| $O$                    | Obstacle clearance score                                        | dimensionless                                |
| $P_{\text{fat}}$       | Probability of fatality given impact                            | dimensionless                                |
| $P_{\text{pop}}$       | Local population density                                        | $\text{persons}/\text{km}^2$                 |
| $R$                    | Ground risk score (public safety)                               | dimensionless                                |
| $S$                    | Surface quality or slope score                                  | dimensionless                                |
| $T$                    | Thrust                                                          | $\text{N}$                                   |
| $V$                    | Airspeed (velocity relative to the air mass)                    | $\text{m}/\text{s}$                          |

| Symbol      | Description                                                          | Units         |
|-------------|----------------------------------------------------------------------|---------------|
| $V_z$       | Vertical velocity (sink rate when positive)                          | m/s           |
| $V_{z,max}$ | Maximum permitted sink rate                                          | m/s           |
| $W$         | Wind alignment score                                                 | dimensionless |
| $w_i$       | Scalar weight on the $i^{\text{th}}$ criterion in the MCDM objective | [0, 1]        |
| $x_{max}$   | Maximum landing-gear stroke                                          | m             |
| $x, y$      | Inertial horizontal position coordinates                             | m             |
| $z$         | Vertical position coordinate, positive downward                      | m             |
| $\eta$      | Propulsion-efficiency factor under failure                           | [0, 1]        |
| $\varphi$   | Rotor tilt angle                                                     | rad           |
| $\chi$      | Course angle                                                         | rad           |
| $\psi$      | Heading angle                                                        | rad           |

# 1. Introduction

## 1.1 Background and Motivation

Autonomous UAVs are increasingly used for cargo logistics, raising safety concerns for operations beyond visual line of sight (BVLOS) over populated terrain. When a mid-flight failure prevents return to a planned recovery site, an emergency landing must be executed autonomously without compromising third-party safety. This is especially critical for heavy-lift platforms, where impact energy from an uncontrolled descent far exceeds the 33.9 J harmlessness threshold below which an impact carries negligible risk of serious injury to an unprotected adult (la Cour-Harbo, 2017), and at 2500 kg even a 1 m/s touchdown carries 1.25 kJ, roughly 37 times this threshold. UK CAA airspace-incident reporting records 1,168 unmanned-aircraft occurrences in 2023, a growing fraction involving BVLOS or autonomous operations (UK CAA, 2024). Reliable autonomous emergency landing is essential for BVLOS heavy-lift UAVs, ensuring failures remain safe and third-party risk is bounded as payload increases.

This dissertation designs and evaluates a guidance system for emergency landing of a hybrid tilt-rotor cargo UAV. The vehicle, developed in the preceding group design phase (Nasar et al., 2025), targets a 500 kg payload over a 200 km mission with VTOL capability. A maximum take-off mass of approximately 2500 kg places it within the UK CAA Specific category, requiring As Low As Reasonably Practicable (ALARP) demonstration of ground-risk mitigation. From 23 April 2025, new operational authorisations follow UK SORA, which replaced the legacy CAP 722A Operating Safety Case route; existing CAP 722A authorisations remain valid until expiry (UK CAA, 2024; UK CAA, 2025). VTOL rules out fixed-wing glide recovery, and the vehicle mass limits passive systems such as parachutes.

Existing approaches fall into three categories. Hardware-based systems absorb impact energy but cannot control landing location (Al-Madani et al., 2024). Perception-driven methods identify suitable terrain but often ignore reachability constraints (Cesetti et al., 2010). Optimisation-based planners (e.g. MPC, RRT, GPM) generate feasible trajectories but are typically validated under nominal conditions rather than failure states. As a result, no approach fully integrates site selection, trajectory generation, and safety enforcement under degraded operation.

## 1.2 Research Question and Objectives

This work addresses that gap by framing emergency landing as a constrained decision-making problem at the guidance layer. A multi-criteria model ranks candidate sites, a kinematic generator produces a reference trajectory, and a runtime safety monitor limits descent rate, lateral error, and approach energy using bounds derived from an impact-energy budget. The control layer is treated as a validated boundary condition, with a PX4-class Auterion Skynode autopilot executing high-level waypoints.

The primary research question is:

*Can a constraint-based guidance architecture, integrating multi-criteria site selection with a runtime safety monitor derived from a landing-gear impact-energy budget, provide measurable third-party safety gains over perception-led and reactive approaches when a heavy-lift cargo UAV suffers a mid-mission propulsion failure?*

The following objectives were pursued:

1. Formulate emergency landing as a multi-criteria decision problem using AHP-derived weights with a dominant public-safety criterion aligned to CAP 722A.
2. Derive a guidance-level safety envelope from landing-gear stroke and stiffness, rather than heuristic rotorcraft thresholds, and link the resulting impact-energy limit to site-selection.
3. Develop a runtime safety monitor that enforces constraints, logs clipping events, and triggers re-scoring when limits are exceeded.
4. Extend the envelope with failure-aware updates so the reachable site set contracts conservatively under propulsion degradation, mass variation, communication loss, and cross-wind disturbance.
5. Validate the pipeline using a 200-run Monte Carlo simulation of a cruise-stage propulsion failure, and benchmark performance against both an unsupervised waypoint baseline and the selection criterion of Cesetti et al. (2010).

### **1.3 Individual Contributions and Novelty**

The author's contributions include: formulating emergency landing as a constrained guidance pipeline; developing a six-criterion weighted-sum MCDM model with AHP weights aligned to NASA heritage and CAP 722A ALARP principles; deriving a guidance-level safety envelope from landing-gear physics rather than generic rotorcraft limits; implementing the runtime safety monitor in MATLAB; designing failure-aware constraint updates linking propulsion, mass, and communication degradation to both the runtime clamp and scoring layer; conducting 200-run Monte Carlo validation with sensitivity analysis; and benchmarking against the Cesetti et al. (2010) vision-based approach. The integrated framework, assessed against UK CAA CAP 722/722A for a 2500 kg vehicle, forms the main novelty and is directly relevant to operators developing Operational Safety Cases for BVLOS cargo missions.

The remainder of this report is organised as follows: literature review (Chapter 2), architecture (Chapter 3), guidance components (Chapter 4-6), failure-aware adaptation (Chapter 7), validation (Chapter 8), regulatory considerations (Chapter 9), and conclusions (Chapter 10).

## **2. Literature Review**

### **2.1 Hardware-based recovery systems**

Ballistic parachutes, airbags, crushable structures, and energy-absorbing landing gear represent the earliest line of UAV safety mitigation. Al-Madani et al. (2024) shows that while these systems can reliably reduce descent velocity and absorb impact energy, they offer no control over where the vehicle lands. For VTOL platforms above several hundred kilograms, required parachute deployment altitudes are often incompatible with the low-altitude failure scenarios most relevant here. As a result, they function primarily as a last-resort terminal mitigation rather than a complete solution.

## 2.2 Perception-driven guidance and safe-site detection

Perception-based methods use onboard cameras or lidar to identify flat, unoccupied terrain. Cesetti et al. (2010) combine SIFT feature tracking with optical-flow variance (threshold 50) for surface classification on a 9 kg helicopter. More recent work (Hua et al., 2026) uses semantic segmentation and multimodal models, incorporating context such as proximity to people. However, these methods assume sites are reachable, prioritise geometric flatness over regulatory risk, and are typically validated on smaller platforms. The group-stage prototype (Nasar et al., 2025) falls within this class; limitations are discussed in Chapter 3.3.

## 2.3 Trajectory planning and fault-tolerant control

Emergency trajectory generation uses sampling-based planners (RRT), MPC for constraint-aware tracking, and optimal vcontrol methods such as the Gauss pseudospectral method (GPM). Nonlinear programming formulations allow explicit enforcement of sink-rate and attitude constraints for fixed-wing VTOL (Al-Madani et al., 2024). Pang et al. (2023) incorporate third-party risk via a Min-cost A\* planner with AHP-derived weights, reducing expected fatalities. At the control level, fault-tolerant methods using gain scheduling and adaptive allocation maintain stability after actuator failure (Yu et al., 2014; Alhosani and Whidborne, 2025). However, these approaches focus on tracking performance rather than public-safety outcomes and do not address landing-site selection.

## 2.4 Multi-Criteria Landing-Site Selection

Multi-criteria decision-making (MCDM) offers a structured way to select between sites with competing attributes. The Weighted Sum Model (WSM) is simple transparent method: candidates are scored across defined criteria, values normalised, and a weighted sum determines preferred option (Triantaphyllou, 2000). In emergency landings, Bayesian MCDM approaches based on three NASA-derived criteria—population risk, reachability, and safe-landing probability—are established. Weights are obtained using Analytic Hierarchy Process (AHP), which checks pairwise consistency and is applied in UAV decision-making (Hamrouni and Eren, 2020).

## 2.5 Gap-analysis and framing of the present work

Across all four strands, emergency landing is treated as separate components, perception, site-selection, planning, and control, each validated independently. Their interactions, particularly whether a selected site is reachable under degraded dynamics and within controller-imposed impact limits, are rarely addressed. Few studies incorporate regulatory ground-risk constraints in real-time guidance loop robust to communication loss, propulsion degradation, or crosswind. This dissertation closes that gap at the guidance layer by integrating: a transparent weighted-sum site-selection model (NASA three-criterion framework, extended with wind, obstacle, and energy terms); a runtime safety monitor enforcing vehicle and impact-energy envelopes; and explicit failure-aware constraint updates. The contribution is an integrated guidance pipeline combining MCDM site-selection with runtime safety monitoring for the 2500kg class.

## 3. Proposed System Architecture

### 3.1 Scope and assumptions

The proposed system focuses on the guidance layer only. It assumes a triple-redundant PX4-class flight control system on Auterion Skynode-class hardware, consistent with the group-stage FCS architecture (Nasar et al., 2025). The inner-loop attitude and rate controllers, Extended Kalman Filter (EKF) state estimation, and mixer/actuator allocation are treated as validated components that track position, velocity, and heading setpoints reliably (Meier et al., 2015). A companion computer hosts the guidance logic while perception outputs, risk maps, and population data are inputs.

Three assumptions define the vehicle envelope: instantaneous mass between 1800 and 2500 kg; available thrust degraded to a fraction  $\eta$  of nominal under partial propulsion failure, with  $\eta \in [0.5, 1.0]$  considered recoverable; and winds up to 15 m/s at approach altitude (Alhosani and Whidborne, 2025)

### 3.2 Guidance pipeline

On receiving an emergency trigger, the system executes a six-stage pipeline:

- (1) perception and state aggregation (terrain map, weather, population, obstacles)
- (2) candidate generation within a reachable radius
- (3) multi-criteria scoring based on reachability, ground risk, obstacle clearance, surface quality, wind alignment, and energy feasibility (Chapter 4)
- (4) site selection (minimum-cost candidate)
- (5) constraint-based trajectory generation with a runtime safety monitor (Chapter 5)
- (6) handoff of waypoint, velocity, and heading commands to the autopilot via MAVLink offboard setpoints

A feedback loop provides updated state, energy, and health information at each guidance cycle, allowing the pipeline to re-score sites if the reachable set changes. The overall architecture is shown in Figure 1.

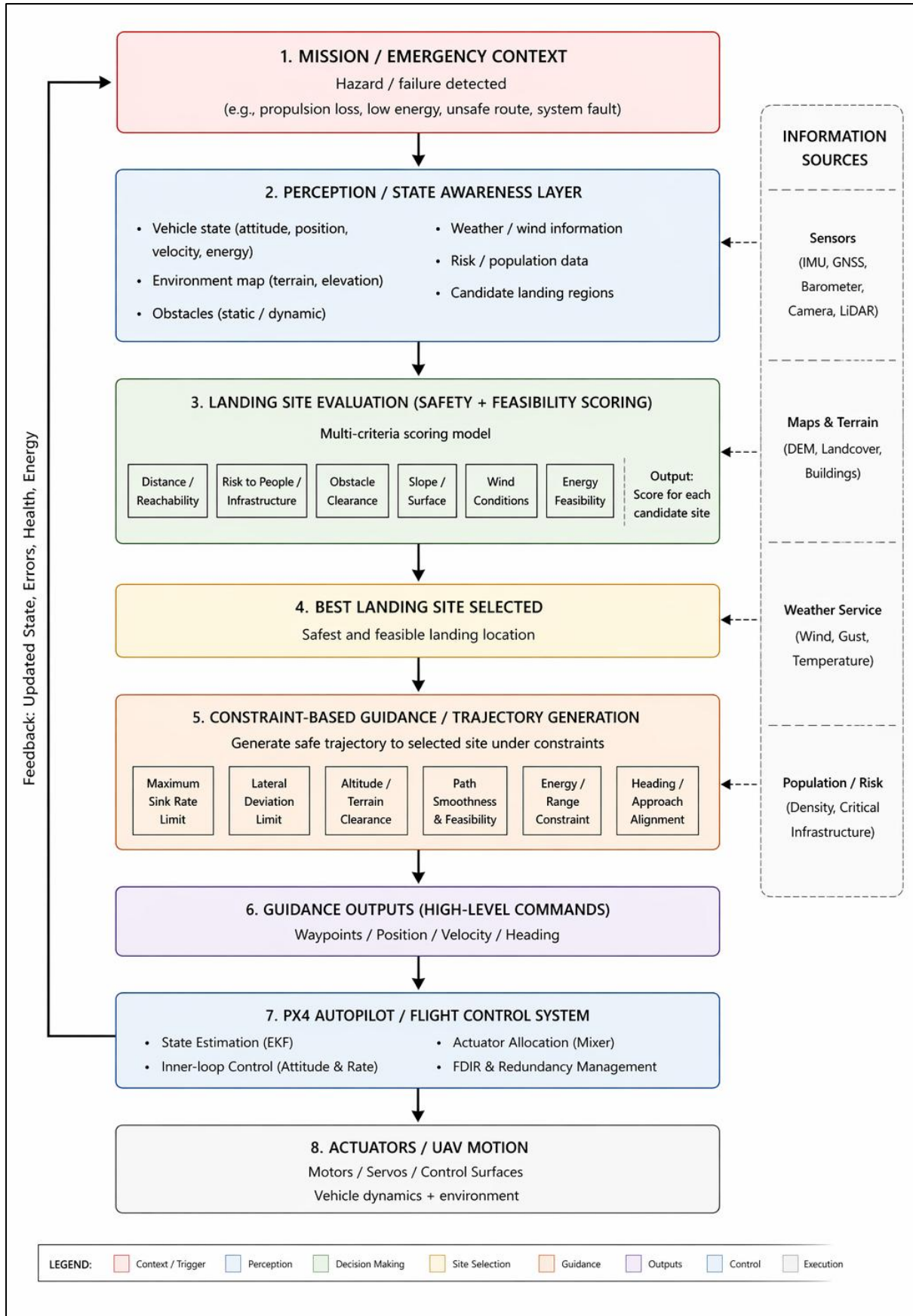


Figure 1: Eight-block guidance architecture showing information flow from perception through site selection, constraint-based trajectory generation, and autopilot execution. The dashed feedback path carries updated state, health, and energy information. DEM stands for the Digital Elevation Model.

### 3.3 Development from the group-level prototype

The present system builds directly on the group-stage prototype (Nasar et al., 2025).

#### 3.3.1 The group-stage prototype

The earlier system implemented a three-stage Detect–Segment–Evade pipeline (Figure 2): YOLOv8 detected airborne obstacles, a SegFormer-style head classified flyable airspace, and an evade module emitted a lateral command (MOVE LEFT / RIGHT / CLEAR) which was passed to the PX4 Auterion Skynode autopilot via MAVLink. Bird detection was validated at 0.93 confidence (Figure 3) under three static test cases.

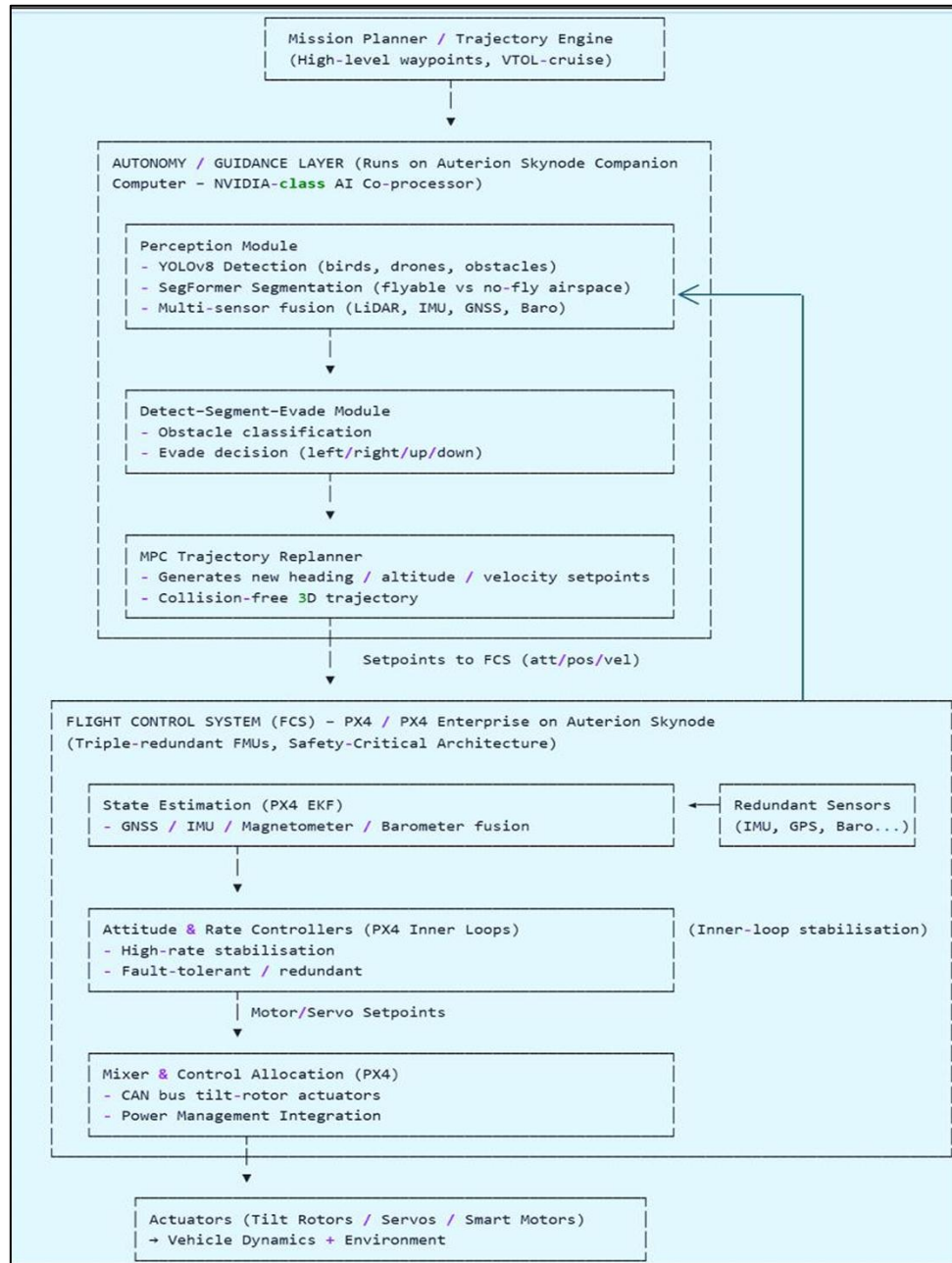


Figure 2: Group-stage perception pipeline (Nasar et al., 2025). A reactive, frame-by-frame architecture: perception (YOLOv8 + SegFormer) detects obstacles and segments the sky, the Detect-Segment-Evade module emits a lateral command, and an MPC replanner translates it into setpoints for the PX4 autopilot. No global landing plan, no safety envelope, no failure awareness.



Figure 3: Group-stage perception test case (Nasar et al., 2025). Left: input image with an airborne obstacle. Right: YOLOv8 detection (confidence 0.93), sky segmentation overlay, and the emitted lateral command: MOVE LEFT / RIGHT, OBSTACLE AHEAD. The system reacts to what is visible in the current frame but cannot plan a global safe landing.

### 3.3.2 Why the prototype is inadequate for emergency landing

Three structural weaknesses make it unsuitable for emergency landing. It is frame-reactive: commands are generated from the current image only, with no model of where the vehicle should ultimately land, it can avoid a bird but cannot answer “where should I set down given a propulsion failure?”. Safety is implicit rather than enforced: if segmentation misclassifies a car park as flyable, the MPC replanner can still command descent rates outside the structural envelope. The pipeline assumes nominal dynamics, generating setpoints as if the aircraft were healthy and ignoring degraded thrust, mass variation, or reduced energy. For a 2500kg vehicle under CAP 722, none of these is acceptable.

### 3.3.3 What the present system adds

The development retains what worked and replaces what did not. The perception stack (YOLOv8 + segmentation) is kept as a Stage 1 input to the new pipeline (Figure 1), and the PX4 autopilot and MAVLink interface are retained. Four components are introduced between perception and the flight control system: candidate generation (covering all reachable regions, not just those currently visible), MCDM scoring, a runtime safety monitor, and failure-aware constraint updates. Table 1 summarises the transition:

#### Development from Group Prototype to Proposed Guidance System:

Table 1: Development from the group-stage machine-learning prototype (Figure 2) to the constraint-based guidance system (Figure 1). Perception (YOLOv8) and the autopilot are retained, while the decision and trajectory layers are replaced with global, safety-enforcing, failure-aware logic. YOLOv8 now acts as one input to the MCDM layer rather than the sole decision-maker.

| Layer          | Group prototype (Figure 2)                   | Present system (Figure 1)             |
|----------------|----------------------------------------------|---------------------------------------|
| Perception     | YOLOv8 + SegFormer on camera feed            | Unchanged (retained as Stage 1 input) |
| Decision logic | Reactive evade module → LEFT / RIGHT command | MCDM scoring over candidate sites     |

| Layer                 | Group prototype (Figure 2)           | Present system (Figure 1)                |
|-----------------------|--------------------------------------|------------------------------------------|
| Trajectory generation | MPC replanner with no envelope check | Pure pursuit with runtime safety monitor |
| Failure handling      | Not represented                      | Failure-aware constraint updates         |
| Autopilot interface   | MAVLink to PX4 Auterion Skynode      | Unchanged                                |
| System behaviour      | Reactive, perception-led             | Proactive, constraint-enforcing          |

## 4. Landing Site Selection Model

### 4.1 Formulation as a weighted-sum MCDM problem

Emergency landing is framed as a discrete MCDM problem over a finite set of candidate sites  $\{s_1, \dots, s_n\}$ . For each site, six criteria are evaluated: reachability  $D$ , ground risk  $R$ , obstacle clearance  $O$ , surface quality  $S$ , wind alignment  $W$ , and energy feasibility  $E$ . The selected site minimises a weighted sum of the normalised criteria:

$$J(s_k) = w_D \hat{D}_k + w_R \hat{R}_k + w_O \hat{O}_k + w_S \hat{S}_k + w_W \hat{W}_k + w_E \hat{E}_k \quad (1)$$

where each  $\hat{c}$  is defined so that lower is better (larger-is-better criteria  $O$  and  $E$  are inverted per equation 2b). WSM is selected over TOPSIS (less transparent audit trail) and AHP-Bayesian (Ramirez-Atencia et al., 2024) (unjustified distributional assumptions) because each  $\hat{c}_i$  contribution to  $J$  is directly inspectable, the  $O(N \cdot 6)$  cost runs at 20 Hz on Skynode, and AHP integrates at weight elicitation (Triantaphyllou, 2000; Hamurcu and Eren, 2020)

### 4.2 Criterion definitions

Reachability  $D_k$  is the minimum energy required to reach a candidate site under the degraded envelope. To first order,  $D_k \approx mg\Delta h + \frac{1}{2}mV^2 \frac{d_k}{L/D}$ , where  $d_k$  is planar distance and  $(L/D)$  is the best glide ratio. Under normal thrust this reduces to a geometric distance measure.

Ground risk  $R_k$  is the expected third-party fatality count for a crash at site  $s_k$ . Following MITRE (2018) and Pang et al. (2023),  $R_k \approx P_{pop}(s_k) \cdot A_{imp} \cdot P_{fat}$ , where  $P_{fat}$  follows the MITRE kinetic-energy fatality curve with a 50% fatality probability at  $E_0 \approx 110J$ , for an unprotected adult. For a 2500kg vehicle this threshold is exceeded at any realistic sink rate, so population density dominates rather than energy reduction

Obstacle clearance  $O_k$  is the approach-corridor distance to the nearest static or dynamic obstacle (from the perception stack). Surface quality  $S_k$  combines slope and land-cover class (Hua et al., 2026) and is the only criterion using vision inputs. Wind alignment  $W_k$  rewards headings aligned with the local wind vector. Energy feasibility  $E_k$  is binary: sites with  $E_{required} > E_{available}$  are removed from the candidate set.

### 4.3 Normalisation and weight elicitation

Each raw criterion is min-max normalised so that  $\hat{c} \in [0, 1]$ , ensuring all terms are comparable:

$$\hat{c}_k = \frac{c_k - c_{\min}}{c_{\max} - c_{\min}} \quad (2a)$$

Weights are obtained using AHP pairwise comparison (Hamurcu and Eren, 2020). Public safety is treated as the dominant criterion, consistent with the NASA three-criterion framework (risk to civilian population, reachability, probability of safe landing) summarised in Ramirez-Atencia et al. (2024), in which risk to the civilian population outweighs vehicle preservation. The weights used are  $w_R = 0.35$ ,  $w_E = 0.25$ ,  $w_D = 0.15$ ,  $w_O = 0.10$ ,  $w_S = 0.10$ , and  $w_W = 0.05$ , with a consistency ratio below 0.10. These weights encode the NASA three-criterion priority of public safety over vehicle preservation; sensitivity to weight selection is examined in Section 8.6.

#### 4.4 Worked example

Three candidate sites are considered for a failure over semi-rural terrain. Site A is the nearest, (a farm track, low population, some cabling). Site B (a cricket pitch, open but located next to a clubhouse with around 80 people). Site C is an open field at the edge of the reachable set, with no population but some crosswind misalignment.

##### Raw Criterion Values for Three Candidate Sites:

*Table 2. Raw criterion values for the three candidate sites in the worked example.*

| Criterion (raw)                    | Site A (track) | Site B (pitch) | Site C (field) |
|------------------------------------|----------------|----------------|----------------|
| Distance, D (m)                    | 320            | 560            | 1050           |
| Risk, R (persons/km <sup>2</sup> ) | 40             | 180            | 5              |
| Obstacle clearance, O (m)          | 6              | 25             | 30             |
| Surface slope, S (°)               | 4              | 1              | 2              |
| Wind offset, W (°)                 | 10             | 15             | 55             |
| Energy margin, E (kJ)              | 2400           | 1900           | 400            |

Applying min-max normalisation (inverting O and E) and the AHP weights, the composite scores are:

$$\hat{c}_k = \frac{c_{\max} - c_k}{c_{\max} - c_{\min}} \quad (2b)$$

##### Normalised Scores and Weighted Composite Cost J:

*Table 3. Normalised scores and weighted composite cost J for the three candidate sites. Site A (J = 0.270) is selected. Values are computed in run\_simulation.m using min-max normalisation with inversion on the obstacle-clearance and energy-margin criteria.*

| Criterion            | Site A | Site B | Site C |
|----------------------|--------|--------|--------|
| $\hat{D}$ (w = 0.15) | 0.000  | 0.329  | 1.000  |

| Criterion                                        | Site A       | Site B       | Site C       |
|--------------------------------------------------|--------------|--------------|--------------|
| $\hat{R}$ ( $w = 0.35$ )                         | 0.200        | 1.000        | 0.000        |
| $\hat{O}$ ( $w = 0.10$ )                         | 1.000        | 0.208        | 0.000        |
| $\hat{S}$ ( $w = 0.10$ )                         | 1.000        | 0.000        | 0.333        |
| $\hat{W}$ ( $w = 0.05$ )                         | 0.000        | 0.111        | 1.000        |
| $\hat{E}$ ( $w = 0.25$ )                         | 0.000        | 0.250        | 1.000        |
| $J = \sum w_i \cdot \hat{c}_i$ (lower is better) | <b>0.270</b> | <b>0.488</b> | <b>0.483</b> |

Site A is selected despite cabling proximity, as low population density dominates while remaining energy-feasible. Site B, although geometrically favourable, is rejected due to the nearby clubhouse ( $\hat{R} = 1.0$ ). Site C is excluded at the edge of the energy-feasible set, with crosswind misalignment. The framework prioritises public safety over geometric convenience, rejecting sites a perception-only approach would accept.

## 5. Constraint-Based Trajectory Generation

### 5.1 Vehicle model

Guidance is based on a 3-DOF point-mass kinematic model, commonly used for UAV outer-loop design (Alhosani and Whidborne, 2025). The model captures the translational motion relevant to trajectory generation while abstracting the attitude dynamics handled by the autopilot:

$$\dot{x} = V \cos \chi, \quad \dot{y} = V \sin \chi, \quad \dot{z} = V_z \quad (3)$$

$$V_z = (mg - T \cos \phi) / m \quad (4)$$

Here,  $V$  is airspeed,  $\chi$  is course angle,  $V_z$  is vertical speed (positive down),  $T$  is available thrust, and  $\phi$  is the rotor tilt angle measured from vertical. Under partial propulsion failure,  $T$  is scaled by  $\eta$ . Faster attitude dynamics are handled by the inner loop, and wind enters as an additive disturbance on  $(\dot{x}, \dot{y})$ . The model is sufficient to verify safety-envelope behaviour while remaining simple enough for the monitor to run at each guidance step on the Skynode companion computer.

### 5.2 Baseline guidance law

A pure-pursuit lateral law with proportional vertical tracking is used as the baseline for a selected site ( $x_r, y_r, z_r = 0$ ). Equation (5) is the pure-pursuit course law in its simplest form, in which the look-ahead point coincides with the target waypoint (Beard and McLain, 2012). The commanded course angle is obtained from the line-of-sight vector between the current position  $(x, y)$  and the reference position  $(x_r, y_r)$ . This vector is given by  $(x_r - x, y_r - y)$ , and the corresponding heading angle is computed using the four-quadrant arctangent function:

$$\chi_{\text{cmd}} = \text{atan2}(y_r - y, x_r - x) \quad (5)$$

The use of the atan2 function ensures the correct heading is obtained over the full 360° range, avoiding quadrant ambiguity. This formulation therefore directs the velocity vector straight toward the selected landing site, representing the limiting case of pure-pursuit guidance with zero look-ahead distance.

The proportional vertical tracking law is derived by defining the altitude error as  $e_z = z - z_r$ , where  $z$  is the current altitude and  $z_r$  is the reference altitude. For a fixed landing site,  $z_r$  is constant, so  $\dot{z}_r = 0$ . Differentiating the altitude error therefore gives  $\dot{e}_z = \dot{z} - \dot{z}_r = \dot{z}$ . Since the vertical velocity is  $V_z = \dot{z}$ , the error dynamics can be written as  $\dot{e}_z = V_z$ . To ensure the altitude error decays smoothly and stably, first-order error dynamics are imposed as  $\dot{e}_z = -k_z e_z$ , where  $k_z > 0$ , which is the standard requirement for exponentially stable tracking in a single-input single-output system (Stevens, Lewis and Johnson, 2016). Equating this with  $\dot{e}_z = V_z$  gives the commanded vertical velocity as  $V_{z,\text{cmd}} = -k_z e_z$ . Substituting  $e_z = z - z_r$  gives:

$$V_{z,\text{cmd}} = -k_z(z - z_r) \quad (6)$$

The descent rate is proportional to altitude error, reducing smoothly as the UAV approaches the landing altitude. Lateral error is  $e_{xy} = \sqrt{(x - x_r)^2 + (y - y_r)^2}$ . Pure pursuit is chosen over L1 or vector-field guidance as it avoids look-ahead tuning under degraded conditions, converges smoothly to the target, and integrates directly with the safety clamp.

### 5.3 Command constraints

The baseline law alone provides no safety guarantees. For a heavy-lift vehicle, a single command outside the envelope can cause structural damage. The following constraints are therefore enforced as hard bounds on the commanded inputs:

$$\begin{aligned} |V_{z,\text{cmd}}| &\leq V_{z,\text{max}} \\ |\phi_{\text{cmd}}| &\leq \phi_{\text{max}} \\ e_{xy} &\leq e_{\text{max}} \\ h(t) - h_{\text{terrain}}(x(t), y(t)) &\geq h_{\text{min}} \\ \frac{1}{2}mV_z(t)^2 &\leq E_{\text{max}} \end{aligned} \quad (7)$$

Equation (7) defines a constraint set rather than a single derived relation, with each bound representing a distinct physical or operational limit.

The sink-rate bound  $V_{z,\text{max}} = 2.1 \text{ m s}^{-1}$  is taken from the Section 6.1 derivation under degraded thrust ( $\eta = 0.7$ ). The terminal energy bound is its equivalent energy form,  $\frac{1}{2}mV_z^2 \leq \frac{1}{2}mV_{z,\text{max}}^2 \equiv E_{\text{max}}$ , giving  $E_{\text{max}} = 5.5 \text{ kJ}$ . It is retained because it expresses the touchdown impact limit in energy terms (the form used to derive  $V_{z,\text{max}}$  in Section 6.1), and is enforced implicitly by the sink-rate clip in Section 6.2.

The tilt bound  $\phi_{\text{max}} = 15^\circ$  preserves rotor thrust authority during the transition between hover and cruise, where the rotors must continue to provide a vertical force component sufficient to maintain altitude under degraded propulsion. The value is set as a conservative engineering choice for the present guidance layer.

The lateral bound  $e_{\max} = 5$  m provides margin over expected pure-pursuit tracking error and defines the valid corridor for obstacle evaluation. The clearance bound  $h_{\min}$  is enforced relative to the digital elevation model used by the EKF altitude estimator (Meier et al., 2015).

Each bound relies on an underlying assumption. The sink-rate and energy limits assume the lumped-mass impact model from Section 6.1. The tilt bound assumes static thrust equilibrium. The lateral bound assumes the perception map remains valid within the defined corridor width. The clearance bound assumes the DEM and EKF altitude estimate are reliable. Together, these assumptions define the limits within which the safety envelope remains valid.

## 6. Safety Envelope

### 6.1 Deriving the sink-rate limit from an impact-energy budget

The sink-rate envelope  $V_{z,\max}$  is derived from first principles of energy conservation at touchdown, modelling the landing gear as a lumped mass–spring–damper system (Figure 4) (Al-Madani et al., 2024).

The physical modelling is intentionally simplified to a one-dimensional vertical energy balance. This is appropriate since: the objective is a single scalar bound  $V_{z,\max}$  enforceable by the runtime safety monitor (Section 6.2) and higher-dimensional models would not yield a more actionable constraint at the guidance layer; horizontal velocity at touchdown is bounded indirectly through the lateral-error and tilt constraints in Equation (7), so the residual horizontal kinetic energy is small relative to the vertical component; and full three-dimensional touchdown modelling, including asymmetric gear response and structural flexibility, lies within structural certification rather than guidance-layer envelope design.

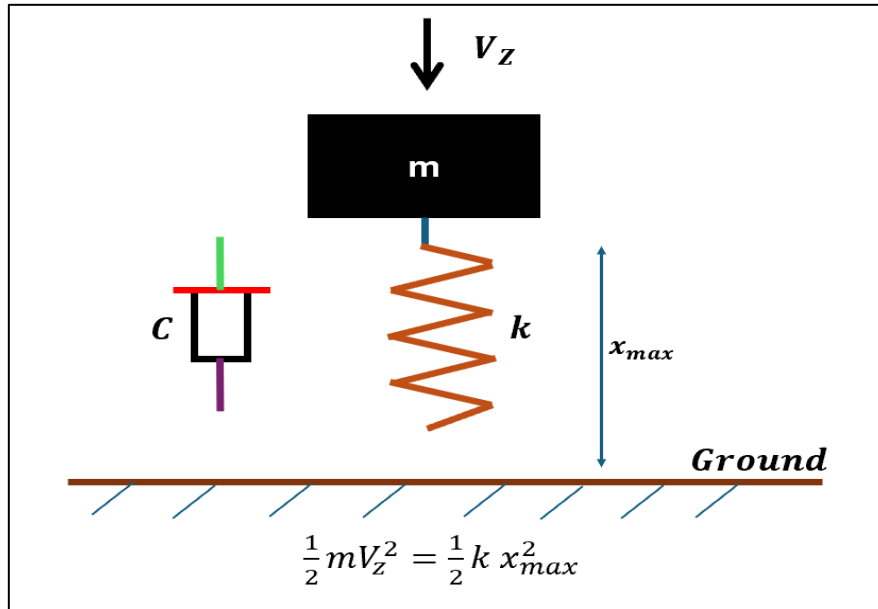


Figure 4: Simplified lumped mass-spring-damper model of the touchdown event used to derive the sink-rate envelope from a specified stroke and effective gear stiffness.

**Step 1:** Kinetic energy at ground contact:

$$KE_{impact} = \frac{1}{2}mV_z^2 \quad (8)$$

**Step 2:** Maximum energy absorbed by the landing gear at full stroke. Damping is neglected as a conservative simplification (the strut velocity is zero at peak compression, so no instantaneous work is done by damping):

$$E_{absorb} = \frac{1}{2}kx_{max}^2 \quad (9)$$

**Step 3:** Safe landing requires  $KE_{impact} \leq E_{absorb}$ . Rearranging gives the envelope used by the safety monitor:

$$V_{z,max} = x_{max} \sqrt{\frac{k}{m}} \quad (10)$$

Maximum landing gear compression ( $x_{max} = 0.30$  m) and gear stiffness ( $k = 0.25$  MN/m) are obtained from landing-gear sizing instead of manufacturer data, as specifications for a 2500 kg tilt-rotor cargo UAV are not publicly available.  $x_{max}$  is estimated using the energy-balance method of Currey (1988) in which the required stroke is set so that the gear absorbs the kinetic energy  $V^2/(2g)$  plus tire-deflection terms at a specified load factor. For a typical sink rate of 3 m/s and load factor  $N = 3$ , this gives stroke ranges of 0.25–0.35 m for vehicles in the 1500–3000 kg class (Dinc et al., 2025), from which the mid-range value of 0.3 m is selected. The stiffness  $k = 0.25$  MN/m is treated as a lumped linear approximation of the nonlinear oleo-pneumatic strut response and is chosen such that, with vehicle mass  $m = 2500$  kg, the resulting  $V_{z,max} = x_{max} \cdot \sqrt{(k/m)} \approx 3.0$  m/s lies within the 1.5–3 m/s normal touchdown range reported for comparable rotorcraft masses, with up to 6 m/s in emergency conditions (Al-Madani et al., 2024). Under reduced thrust,  $V_{z,max}$  is scaled by  $\eta$  as a safety adjustment to reflect reduced ability to arrest descent. This is not derived from the model but imposed conservatively; for  $\eta = 0.7$ ,  $V_{z,max} \approx 2.1$  m/s and  $E_{max} = \frac{1}{2}mV_{z,max}^2 \approx 5.5$  kJ. Sensitivity to these parameters (Section 8.6) shows the  $V_z$  clamp remains within published touchdown limits across  $\pm 30\%$  gear-parameter variation, with stroke length dominating energy.

The damper  $c$  in Figure 4 is included for physical realism but does not appear in equation (10) for the reason given in Step 2. The post-impact oscillation is damped by  $c$  itself: in a real oleo-pneumatic strut, hydraulic fluid forced through metering orifices during compression and rebound dissipates kinetic energy as heat, bringing the airframe to rest within a small number of cycles (Dinc et al., 2025). Damping therefore makes the gear absorb more than  $\frac{1}{2}kx_{max}^2$  in practice, so neglecting it underestimates total absorption and yields a lower allowable sink rate, making equation (10) a conservative bound. The runtime monitor in Section 6.2 clamps  $V_z$  to this spring-only bound, so explicitly modelling  $c$  would only relax the constraint and is unnecessary.

Additional assumptions include (A1) the airframe is rigid above the gear, so all impact energy is absorbed by the landing gear, and (A2) gear stiffness is modelled as linear over the stroke  $0 \leq x \leq x_{max}$ , which underestimates absorbed energy relative to real stiffening behaviour.

## 6.2 Runtime safety monitor

Constraints are enforced by a runtime monitor placed between the baseline guidance law and the autopilot. For each command, the monitor applies a projection (clipping) step that pulls any out-of-envelope inputs back to the allowable boundary:

$$V_{z,cmd} \leftarrow \text{clip}(V_{z,cmd}, -V_{z,max}, +V_{z,max}) \quad (11)$$

$$\phi_{cmd} \leftarrow \text{clip}(\phi_{cmd}, -\phi_{max}, +\phi_{max}) \quad (12)$$

If the clipped command causes lateral error to exceed  $e_{max}$ , or if the required energy along the current trajectory exceeds the available energy ( $E_{required} > E_{available}$ ), the monitor raises an event that triggers re-scoring of candidate landing sites (Chapter 7). This is a clipping-based approximation of a Control Barrier Function (CBF). A more formal CBF formulation could be used in future work, but the transparent clipping strategy is preferred here as it is easy to verify, requires no additional tuning, and guarantees bounded worst-case behaviour. The monitor runs at each guidance update and logs all clipping events, providing auditable evidence in line with UK SORA / CAP 722A risk-assessment requirements.

## 7. Failure-Aware Adaptation

The safety envelope must adapt when vehicle dynamics degrade, as limits set for nominal propulsion may not be achievable under failure. This chapter outlines how constraints are updated in real time across four representative failure modes.

### Failure-Aware Safety Envelope under Degradation Modes:

*Table 4: Failure-aware envelope contraction.  $V_{z,max}$  scales linearly with  $\eta$  under propulsion degradation and with  $m^{-1/2}$  under mass change;  $E_{max}$  follows quadratically. Communication and wind degradations leave the envelope intact but alter the scoring layer. The  $\eta = 0.7$  row corresponds to the cruise-stage propulsion failure validated in Chapter 8.*

| Failure Mode                         | Trigger                                                      | Vz,max (m/s) | E <sub>max</sub> (kJ) | Reachable Set Behaviour                                                    |
|--------------------------------------|--------------------------------------------------------------|--------------|-----------------------|----------------------------------------------------------------------------|
| Nominal                              | None                                                         | 3.00         | 11.25                 | Full candidate set; all sites scored                                       |
| Propulsion ( $\eta = 0.7$ )          | Fault Detection, Isolation, and Recovery (FDIR) thrust ratio | 2.10         | 5.51                  | Sites with $E_{required} > E_{available}$ removed; reachable set contracts |
| Propulsion ( $\eta = 0.5$ )          | FDIR thrust ratio                                            | 1.50         | 2.81                  | Reachable set shrinks to immediate vicinity; ground risk dominates scoring |
| Payload release (m: 2500 to 2000 kg) | Discrete mass event                                          | 3.35         | 11.25                 | Envelope expands by ~12%; energy budget recomputed; sites re-scored        |
| Communication loss                   | Link timeout                                                 | 3.00         | 11.25                 | Operator override disabled; system defaults to highest-ranked candidate    |

| Failure Mode               | Trigger      | $V_{z,max}$<br>(m/s) | $E_{max}$<br>(kJ) | Reachable Set Behaviour                                             |
|----------------------------|--------------|----------------------|-------------------|---------------------------------------------------------------------|
| Sustained crosswind<br>> W | Wind monitor | 3.00                 | 11.25             | Sites with poor wind alignment penalised in W; re-scoring triggered |

## 7.1 Propulsion degradation

Let  $\eta \in [0,1]$  represent the thrust-to-nominal ratio from the FDIR layer. As  $\eta$  decreases, the feasible glide slope steepens. The monitor scales  $V_{z,max}$  to  $\eta \cdot V_{z,max}$ , increasing  $E_{required}$  for any given site. Sites with  $\hat{E} > 1$  (infeasible) are removed and the WSM is re-evaluated. As  $\eta$  approaches 0.5, the reachable set shrinks to the immediate vicinity and ground risk dominates, which is the intended conservative behaviour.

## 7.2 Mass variation and payload release

Mass changes discretely if payload is released. Since  $V_{z,max} \propto m^{-1/2}$ , releasing 500kg from 2500kg expands the envelope by  $\sqrt{2500/2000} \approx 1.12$ . The energy envelope relaxes accordingly. Mass is updated as a live input to the impact-energy model and sites are re-scored after each discrete event.

## 7.3 Communication degradation

Loss of link removes operator override. The system defaults to the highest-ranked candidate, logs the decision, and proceeds. Site selection and trajectory generation run on board, so no real-time link is required, aligning with CAP 722 autonomous contingency requirements.

## 7.4 Wind and environmental disturbances

Wind gusts affect the baseline law through additive terms in  $(\dot{x}, \dot{y})$ . The monitor absorbs moderate disturbances via the clamp. For sustained crosswind exceeding W, a site may become unreachable on a safe heading, prompting re-scoring.

# 8. Validation and Results

## 8.1 Simulation setup

The guidance pipeline is evaluated using the 3-DOF kinematic model from Section 5.1 with parameters from Table 5. A cruise-stage propulsion failure is introduced at  $t = 0$ , at 100m altitude and 25m/s ground speed, with the three candidate sites from Chapter 4. A 10m/s crosswind and additive Gaussian velocity noise ( $\sigma = 0.5$  m/s) are included. Two configurations are compared: a baseline pure-pursuit with proportional descent and no safety monitor, and the full constraint-based pipeline.

**Simulation Parameters for Monte Carlo Validation:***Table 5: Simulation parameters used for the validation run.*

| Parameter                                                              | Value | Unit |
|------------------------------------------------------------------------|-------|------|
| Vehicle mass, $m$                                                      | 2500  | kg   |
| Nominal thrust-to-weight ratio                                         | 1.2   | -    |
| Post-failure thrust ratio, $\eta$                                      | 0.7   | -    |
| Effective gear stiffness, $k$                                          | 0.25  | MN/m |
| Effective gear stroke, $x_{\max}$                                      | 0.30  | m    |
| Max sink rate, $V_{z,\max}$ (nominal)                                  | 3.0   | m/s  |
| Max sink rate, $V_{z,\max}$ (post-failure, $\eta \cdot V_{z,\max}$ )   | 2.1   | m/s  |
| Max tilt, $\phi_{\max}$                                                | 15    | °    |
| Max lateral error, $e_{\max}$                                          | 5     | m    |
| Max impact energy, $E_{\max} = \frac{1}{2} \cdot m \cdot V_{z,\max}^2$ | 5.5   | kJ   |
| Baseline vertical gain, $k_z$                                          | 0.15  | 1/s  |
| Constraint-based vertical gain, $k_z$                                  | 0.08  | 1/s  |
| Guidance update rate                                                   | 20    | Hz   |
| Crosswind speed                                                        | 10    | m/s  |
| Velocity noise standard deviation                                      | 0.5   | m/s  |
| Simulation horizon                                                     | 90    | s    |
| Monte Carlo runs                                                       | 200   | -    |

**8.2 Scenario and results**

Monte Carlo simulations ( $N = 200$ ) were performed with wind-direction uncertainty of  $\pm 30^\circ$  and initial position variation of  $\pm 10$  m. The baseline uses proportional descent ( $k_z = 0.15$ ) and selects Site B, while the constraint-based case uses  $k_z = 0.08$ , applies the safety monitor, and selects Site A via MCDM. The lower gain ( $k_z = 0.08$ ) reduces reliance on safety clipping, while the higher baseline gain ( $k_z = 0.15$ ) represents unconstrained waypoint-following. The 93.4% reduction in peak impact energy reflects the combined effect of envelope-aware tuning and constraint enforcement; the monitor's contribution is isolated in the  $V_{z,\max}$  sweep (Section 8.3)

### Monte Carlo Results for Baseline and Constraint-Based Guidance:

Table 6: Monte Carlo results ( $N = 200$ ) for a cruise-stage propulsion-failure ( $\eta = 0.7$ , 10 m/s crosswind). The constraint-based pipeline eliminates all envelope violations (200/200 to 0/200), reduces peak impact energy by 93.4%, and reduces the mean per-run peak energy by 94.2%. Both laws taper to near-zero  $V_z$  at touchdown via the proportional law's  $z \rightarrow 0$  behaviour; the headline metric is therefore peak descent  $V_z$ , since premature ground contact would impose impact at the in-flight value (15 m/s peak, 281 kJ) rather than the controlled touchdown value. The longer descent time reflects the sink-rate clamp, an intentional safety trade-off for a 2500 kg vehicle.

| Metric                            | Baseline  | Constraint-based | Improvement |
|-----------------------------------|-----------|------------------|-------------|
| Safety-envelope violations (runs) | 200 / 200 | 0 / 200          | 100%        |
| Peak in-flight kinetic energy (J) | 341,312   | 22,531           | 93.4%       |
| Mean peak energy per run (J)      | 298,035   | 17,178           | 94.2%       |
| Mean touchdown $V_z$ (m/s)        | 0.01      | 0.03             | comparable  |
| Mean touchdown $e_{xy}$ (m)       | 0.71      | 0.75             | comparable  |
| Mean time to touchdown (s)        | 45.6      | 89.2             | -           |
| Selected site (modal)             | Site B    | Site A           | -           |

Figures 5–7 confirm the complementary picture: the baseline violates  $V_{z,max}$  for  $\sim 12$  s with a peak above 15 m/s and reaches Site B in  $\sim 46$  s, while the constraint-based law holds the envelope throughout, tracking Site A over  $\sim 89$  s. Both paths curve northward under the 10 m/s crosswind. The 94.2% mean reduction confirms consistent gain rather than outlier-driven.

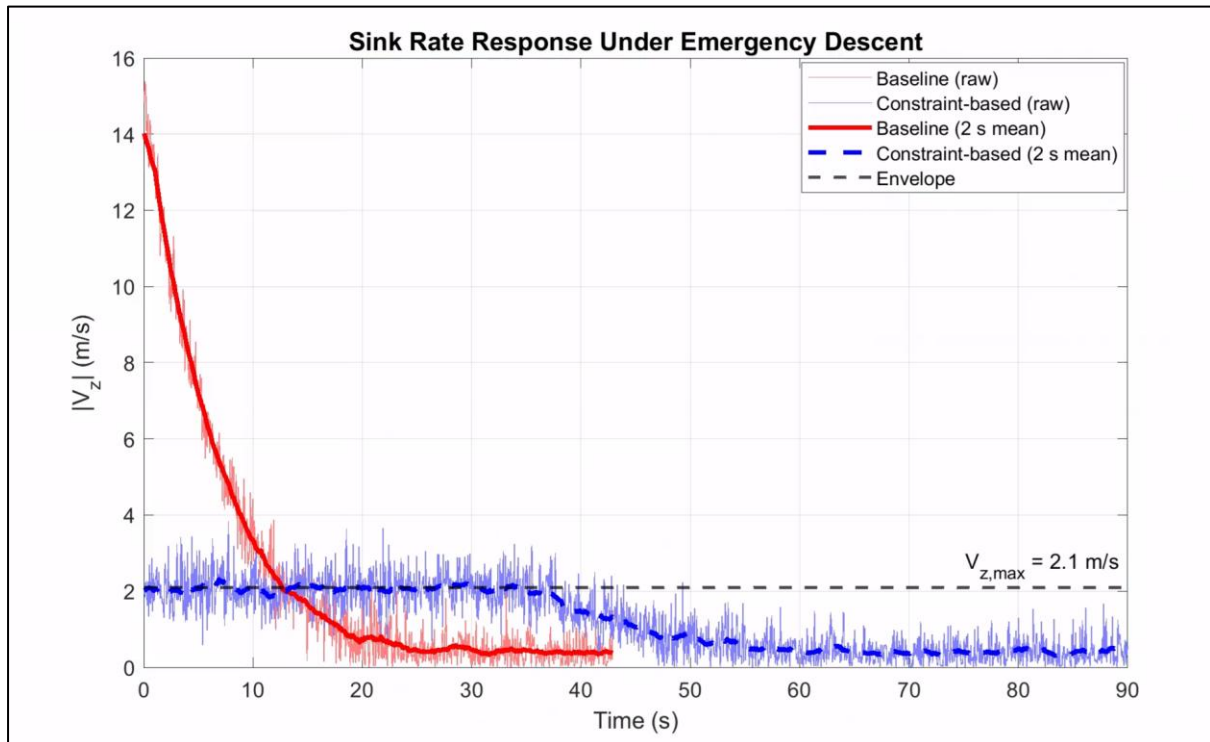


Figure 5: Sink rate  $|V_z|$  against time for a representative run. The baseline (red) violates  $V_{z,max} = 2.1$  m/s for approximately the first 12 seconds, peaking above 15 m/s. The constraint-based law (blue) remains inside the envelope throughout, at the cost of a longer approach time.

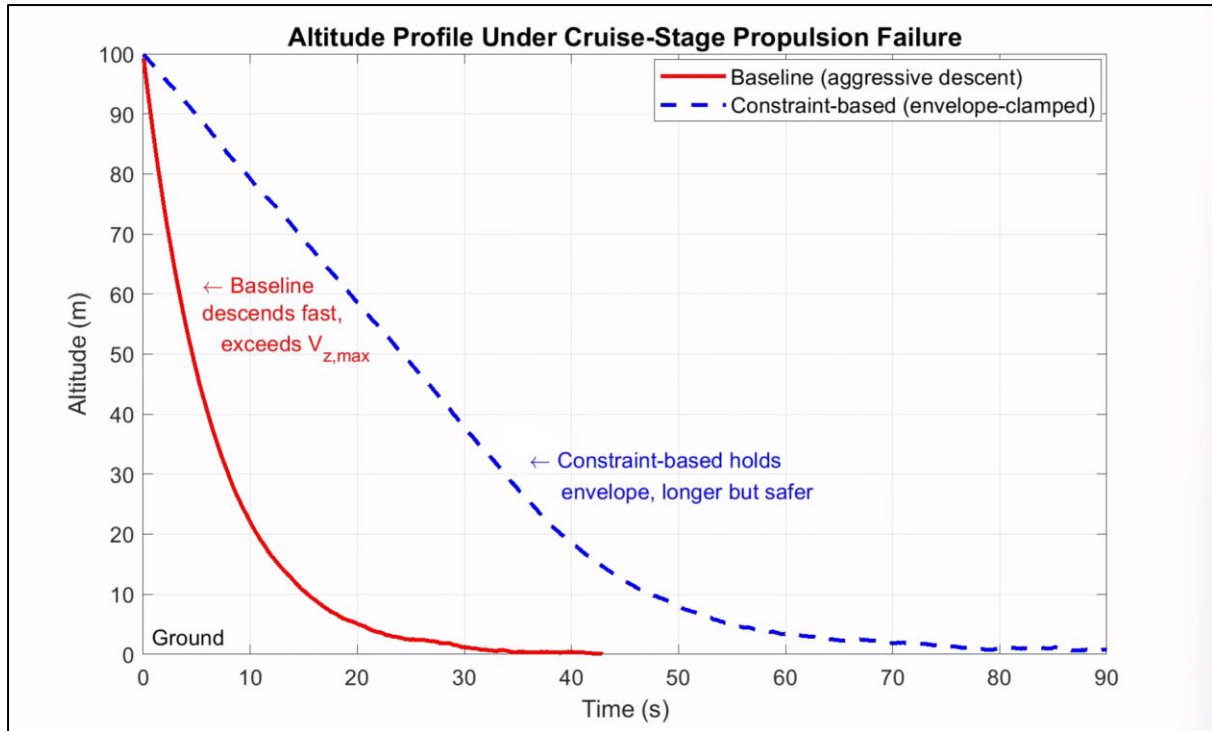


Figure 6: Altitude profile against time. The baseline (red) reaches ground in ~46 s using an unclamped sink rate peaking at 15 m/s (Figure 5), which is catastrophic for a 2500 kg vehicle. The constraint-based profile (blue, dashed) takes ~89 s but remains within the 2.1 m/s envelope throughout.

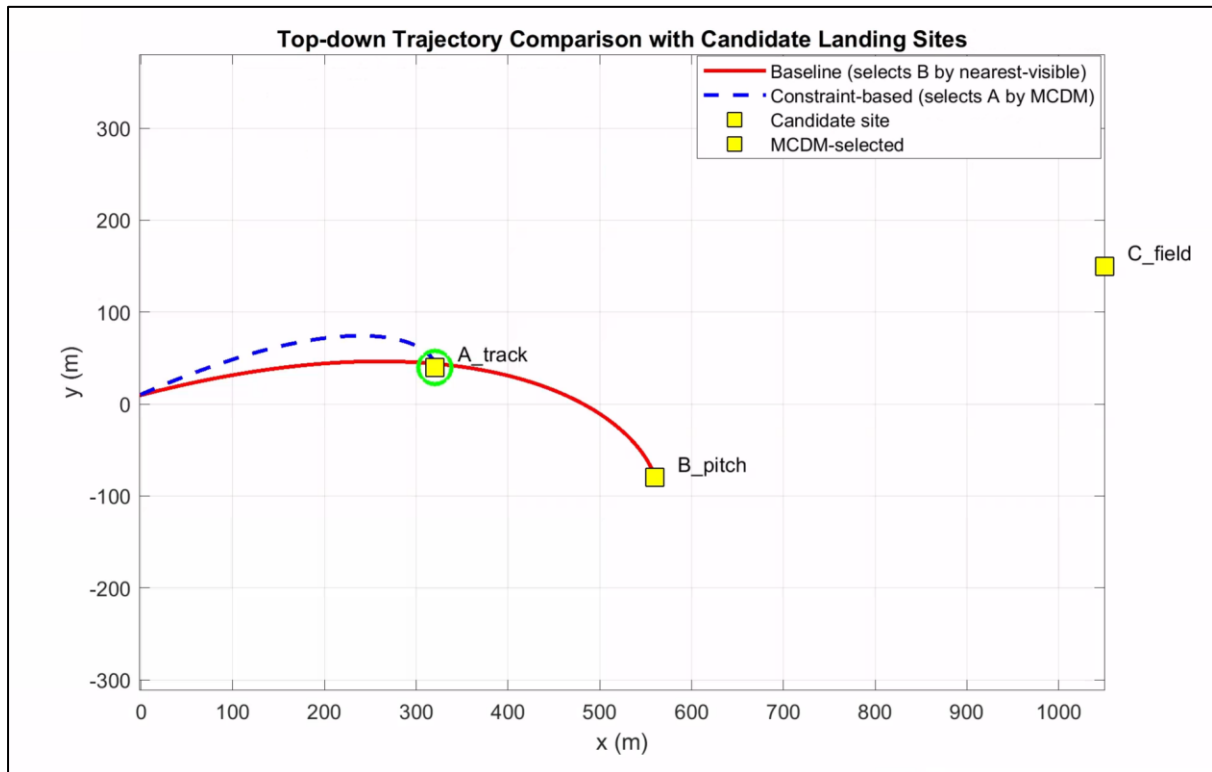


Figure 7: Top-down trajectory. Baseline selects Site B by nearest-visible heuristic and violates the envelope during descent. Constraint-based selects Site A through MCDM (lowest  $J = 0.270$ ), avoiding the clubhouse. Site C is outside the energy-feasible reachable set at  $\eta = 0.7$ . Both tracks curve northward under 10 m/s crosswind.

### 8.3 Sensitivity analysis

The 200-run Monte Carlo fixes  $V_{z,max} = 2.1\text{m/s}$  (the  $\eta = 0.7$  post-failure envelope). To test robustness to assumption changes,  $V_{z,max}$  was swept from 1.5 to 5.0 m/s at fixed  $\eta$ . Each sweep point uses a fresh 200-run Monte Carlo with identical wind, noise, and initial-condition distributions. Results are given in Table 7.

#### Sensitivity Analysis of Sink-Rate Constraint ( $V_{z,max}$ ):

*Table 7: Sensitivity sweep of  $V_{z,max}$  envelope at fixed  $\eta = 0.7$ . Violations remain at zero across the full 1.5 to 5.0 m/s range, confirming that the clip-based safety monitor enforces its bound regardless of the bound value. Peak energy and descent time trade off monotonically with the envelope, illustrating the safety-time Pareto front.*

| $V_{z,max}$ (m/s) | $E_{max}$ (kJ) | Violations | Peak $E$ (kJ) | Mean $E$ (kJ) | Time (s) |
|-------------------|----------------|------------|---------------|---------------|----------|
| 1.5               | 2.8            | 0 / 200    | 19.6          | 12.3          | 90.0     |
| 2.1               | 5.5            | 0 / 200    | 26.0          | 16.8          | 89.1     |
| 3.0               | 11.2           | 0 / 200    | 31.9          | 25.2          | 84.6     |
| 4.0               | 20.0           | 0 / 200    | 52.2          | 37.1          | 82.6     |
| 5.0               | 31.2           | 0 / 200    | 58.3          | 50.0          | 80.2     |

Two insights follow. Violations remain at zero across a  $3.3\times$  range of  $V_{z,max}$ , confirming that the clip-based monitor enforces its bound regardless of its value. The envelope choice introduces a trade-off: tightening from 5.0 to 1.5 m/s reduces peak impact energy by 66% but increases descent time by 12% and shrinks the reachable set. The 2.1 m/s envelope at  $\eta = 0.7$  is therefore a conservative operating point.

A second sweep assesses AHP weight sensitivity. With  $w_O$ ,  $w_S$ , and  $w_W$  fixed,  $w_R$  is varied over  $\{0.25, 0.35, 0.45\}$ , and  $w_D$ ,  $w_E$  are rescaled in a 3:5 ratio (at  $w_R = 0.35$ , the sweep reproduces Table 3:  $J_A = 0.270$ ,  $J_B = 0.488$ ,  $J_C = 0.483$ ). Site A remains preferred in all cases, though its margin over Site C narrows as  $w_R$  increases ( $\Delta J$  from +0.333 to +0.093), reflecting Site C's low population density. The ranking is unchanged across  $\pm 0.10$  variation in the dominant weight, supporting the regulatory-priority claim in Section 10.1.

### 8.4 Validation against established theory

Six elements of the framework are validated against independently published theory and data, as summarised in Table 8. Each entry links a design choice to its source and the level of agreement observed.

**Validation against Established Theory and Benchmarks:**

*Table 8: Validation of framework elements against established theory and published benchmarks. Each design choice is traceable to a primary source, and the observed agreement is either a direct numerical match or within the published uncertainty band.*

| Element                                       | Theory / source                                                                              | Agreement                                                                                            |
|-----------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| $V_{z,\max} = 3.0$ m/s nominal envelope       | Rotorcraft touchdown limits 1.5-3 m/s normal, up to 6 m/s emergency (Al-Madani et al., 2024) | Within published band                                                                                |
| $E_0 = 110$ J fatality threshold in $\hat{R}$ | RCC fatality curve via MITRE sUAS (2018); la Cour-Harbo (2017)                               | Value used directly                                                                                  |
| WSM + AHP weight elicitation                  | Triantaphyllou (2000); Hamurcu and Eren (2020)                                               | Method followed; consistency ratio < 0.10                                                            |
| Pure-pursuit lateral guidance                 | Standard outer-loop theory (Alhosani and Whidborne, 2025)                                    | Measured $e_{xy} \approx 0.7$ m matches published 1-2 m range                                        |
| Weighted ground-risk dominance ( $w_R$ )      | CAP 722 ALARP / NASA three-criterion heritage                                                | $w_R = 0.35$ encodes ALARP; failure-aware logic preserves hierarchy                                  |
| Vision-only baseline comparison               | Cesetti et al. (2010)                                                                        | Structural comparison at $w_S = 1$ ; $J = 0.488$ vs 0.270. Like-for-like reproduction outside scope. |

**8.5 Benchmark against Cesetti et al. (2010)**

Cesetti et al.'s SIFT-based pipeline was demonstrated on a 9 kg helicopter, so a hardware-level reproduction is outside scope. Instead, their selection logic (surface flatness only) is reproduced exactly by setting  $w_S = 1$  and all other weights to zero, isolating decision logic under identical dynamics. The result selects Site B ( $J = 0.488$  vs 0.270 for the proposed framework), yielding the baseline 341 kJ peak energy and 200/200 violations reported in Table 6. The benchmark therefore demonstrates that the runtime monitor is necessary for safety and the multi-criteria scoring is necessary for regulatory acceptability. Neither alone is sufficient.

**8.6 Limitations, trade-offs, and uncertainty**

The framework makes five bounded simplifications. The 3-DOF model excludes attitude dynamics, so inner-loop interactions are not captured. The impact-energy model uses a linear-spring landing gear. Nonlinear stiffening shifts  $E_{\max}$  by up to  $\pm 30\%$  under  $\pm 30\%$  variation in  $k$  (Table 9). Damping is omitted (Section 6.1) as a conservative one-sided simplification. Stroke length  $x_{\max}$  is the dominant sensitivity, with  $\pm 30\%$  variation producing  $-51\%$  to  $+69\%$  change in  $E_{\max}$ , reflecting the quadratic dependence  $E_{\max} = \frac{1}{2}kx_{\max}^2$ . Mass enters only through  $V_{z,\max}$ , leaving the energy bound invariant to mass uncertainty.

The  $V_z$  clamp remains within the 1.5–6.0 m/s touchdown range across all perturbations tested, with  $\eta = 0.5$  on the lower bound. A deterministic 10 m/s crosswind is used, turbulence and

gusts would increase variance. Perception is assumed perfect, with no propagation of detection errors.

These assumptions define a trade-off between safety, descent time, reachability, and model fidelity. Safety and descent time are quantified in Table 7, reachability through the propulsion-degradation analysis in Section 7.1, and fidelity is identified as future work in Chapter 10.

#### Sensitivity of Impact-Energy Envelope to Parameter Variation:

*Table 9: One-at-a-time sensitivity of the impact-energy envelope to parameter perturbations. Monitor functionality is assessed against the 1.5–6.0 m/s normal-touchdown range (Al-Madani et al., 2024).*

| Parameter                    | Nominal   | Range      | $V_{z,max}$ range | $E_{max}$ range      | $\Delta E_{max}$ | Monitor functional?                |
|------------------------------|-----------|------------|-------------------|----------------------|------------------|------------------------------------|
| Gear stiffness $k$           | 0.25 MN/m | $\pm 30\%$ | 2.51 – 3.42 m/s   | 7.88 – 14.63 kJ      | $\pm 30\%$       | Yes                                |
| Stroke $x_{max}$             | 0.30 m    | $\pm 30\%$ | 2.10 – 3.90 m/s   | 5.51 – 19.01 kJ      | –51% / +69%      | Yes                                |
| Vehicle mass $m$             | 2500 kg   | $\pm 10\%$ | 2.86 – 3.16 m/s   | 11.25 kJ (invariant) | 0%               | Yes                                |
| Propulsion efficiency $\eta$ | 0.7       | 0.5 – 0.9  | 1.50 – 2.70 m/s   | 2.81 – 9.11 kJ       | –49% / +65%      | Yes ( $\eta = 0.5$ on lower bound) |

## 9. Regulatory, Ethical, and Environmental Considerations

A 2500 kg UAV falls within the UK CAA Specific category (CAP 722). From 23 April 2025, authorisations follow UK SORA, while CAP 722A Operating Safety Case authorisations remain valid until expiry (UK CAA, 2024; UK CAA, 2025). The guidance system supports this via auditable scoring and clipping logs (ALARP), avoidance of uninvolved persons (1:1 rule, 50 m separation) embedded as a dominant weight on R, and rule-based decision-making.

Ethically, the framework prioritises third-party safety over vehicle preservation ( $w_R = 0.35$ ). This aligns with the Engineering Council’s Statement of Ethical Principles, which holds public safety paramount (Engineering Council, 2017). Sensitivity lies in weight selection; changes in  $w_R$  relative to  $w_D$  can alter routing, so deployment should include stakeholder review (Pang et al., 2023).

Environmentally, trajectories avoid populated and energy-infeasible regions, reducing uncontrolled descent and debris spread. Hybrid-electric propulsion limits emissions, and the guidance layer adds no footprint. Residual risk includes battery or fuel leakage, which could be addressed by adding ecological criteria to the MCDM model.

Residual risks (Table 10) are dominated by concurrent degradations not jointly validated. Single-point propulsion loss is handled via  $V_{z,max}$  contraction. Perception errors are partly mitigated by separating surface and risk criteria, though probabilistic modelling is absent. Landing-gear uncertainty shifts the energy envelope by up to +69%/–51% under  $\pm 30\%$  stroke perturbation (Table 9), but constraints remain enforced. Sensor degradation is handled at the autopilot level, with lateral bounds as backup. Unmapped obstacles remain a residual risk if missed by sensing and prior maps.

### Residual Risks and Mitigation Strategies:

*Table 10: Residual failure modes, qualitative likelihood and severity, and the intended mitigation route. The highest residual risk is the concurrent-degradation case, which is handled channel-by-channel in Chapter 7 but not jointly validated in simulation; this is the principal route identified for future work.*

| Failure Mode                                                     | Likelihood | Severity  | Intended Mitigation                                                                                                                                                                                                  |
|------------------------------------------------------------------|------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single-point propulsion failure ( $\eta$ drop)                   | Moderate   | High      | Handled directly by reducing the allowable sink rate, with $V_{z,max}$ contracting to $\eta \cdot V_{z,max}$ . So, sites that become unreachable or energy-infeasible are removed from the MCDM candidate set.       |
| Concurrent degradations ( $\eta$ + cross-wind + link loss)       | Low        | Very High | Not fully validated as a combined case. Each degradation is handled individually (propulsion, wind, and communication), and joint behaviour is identified as future work with a Monte Carlo analysis proposed.       |
| Perception misclassification (e.g., car park flagged as flyable) | Moderate   | High      | Partially mitigated by separating surface quality and ground-risk criteria, so an over-confident surface estimate is balanced by population risk. A probabilistic perception interface is identified as future work. |
| Landing-gear parameter uncertainty ( $k$ , $x_{max}$ )           | High       | Moderate  | Shifts the energy envelope by up to +69%/-51% under $\pm 30\%$ stroke perturbation (Table 9), with stroke length the dominant sensitivity. The safety monitor remains functional across the tested range.            |
| AHP weight bias favouring particular communities                 | Moderate   | Moderate  | Addressed through structured stakeholder input and sensitivity analysis on weight selection, particularly for the ground-risk term.                                                                                  |
| Gradual sensor degradation (EKF drift, GNSS multipath)           | Moderate   | Moderate  | Handled outside this framework by the autopilot's estimation checks. The lateral-error constraint provides an additional safety bound if state estimates degrade.                                                    |
| Unmapped obstacle in selected approach corridor                  | Low        | High      | Mitigated through real-time perception and re-scoring if lateral deviation occurs. Residual risk remains if the obstacle is missed by both onboard sensing and prior maps.                                           |

Residual risk is concentrated in the concurrent-degradation case (low likelihood, very high severity), which is identified for joint Monte Carlo validation as part of the UK SORA / CAP 722A safety-case evidence package.

## 10. Conclusions and Future Work

This dissertation developed a constraint-based emergency landing guidance system for a 2500 kg heavy-lift tilt-rotor UAV, integrating MCDM site selection with a runtime safety monitor.

### 10.1 Principal conclusions

The investigation aimed to determine whether a constraint-based guidance architecture could provide measurable safety gains over reactive approaches when a heavy-lift cargo UAV suffers a mid-mission propulsion failure. The Monte Carlo results show a clear improvement: 200/200 envelope violations were eliminated, peak impact energy fell by 93.4% (341 kJ to 22.5 kJ), and mean peak energy per-run fell by 94.2%. The sensitivity analysis confirmed that the clipping-based monitor enforces its limits across a  $3.3\times$  range of  $V_{z,max}$ , highlighting a clear safety and descent trade-off at the 2.1 m/s operating point. The sweep also revealed a Pareto trade-off between impact-energy minimisation and descent time, suggesting the envelope should be selected per mission rather than fixed at a single value.

Three deductions follow. First, perception-only selection is insufficient at this mass class: the  $w_S = 1$  benchmark selects Site B and yields 200/200 violations, so geometric flatness alone cannot satisfy ALARP without envelope enforcement. Second, the converse holds: a monitor without weighted scoring would safely guide toward geometrically favourable sites regardless of population, so  $w_R = 0.35$  is what translates envelope compliance into regulatory compliance. Third, deriving  $V_{z,max}$  from  $x_{max}\sqrt{k/m}$  enables consistent envelope contraction under propulsion loss, as the  $\eta$  scaling follows directly from the same relation. The comparison with Cesetti et al. (2010) therefore acts as a decomposition rather than a like-for-like benchmark, isolating the contribution of each component.

### 10.2 Difficulties encountered

The AHP weights were set heuristically rather than through operator input, though Section 8.3 confirms the site ranking is robust to  $\pm 0.10$  variation in  $w_R$ . The impact-energy model relied on scaled landing-gear parameters, as manufacturer data for a 2500 kg tilt-rotor is not publicly available, introducing an estimated  $\pm 30\%$  uncertainty. Finally, validation was limited to a 3-DOF kinematic model, meaning a full 6-DOF re-run remains as future work.

### 10.3 Future work

Three priorities follow: 6-DOF validation (Simulink or PX4 SITL), Hardware-In-The-Loop testing on the Skynode platform with fault injection, and replacement of placeholder AHP weights with operator-informed values. Longer-term work includes probabilistic perception and a CBF formulation for formal safety guarantees.

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## Appendix A: Worked MCDM Calculation

This appendix records the full calculation that produced Tables 2 and 3 in Chapter 4, provided for reproducibility. Min-max normalisation inverts criteria where a larger raw value corresponds to better performance (O, E) so that lower normalised value always corresponds to better site. The weighted composite J is a linear combination of the normalised values. For convenience, equations (2a) and (2b) are restated below:

Step 1: Min-max normalise each criterion. For a criterion where smaller is better (D, R, S, W):

$$\hat{c}_k = \frac{c_k - c_{\min}}{c_{\max} - c_{\min}}$$

For a criterion where larger is better (O, E), invert:

$$\hat{c}_k = \frac{c_{\max} - c_k}{c_{\max} - c_{\min}}$$

Step 2: Compute J for each site:

$$J_k = 0.15 \cdot \hat{D}_k + 0.35 \cdot \hat{R}_k + 0.10 \cdot \hat{O}_k + 0.10 \cdot \hat{S}_k + 0.05 \cdot \hat{W}_k + 0.25 \cdot \hat{E}_k$$

Substituting the raw values from Table 2 gives  $J_A = 0.270$ ,  $J_B = 0.488$ ,  $J_C = 0.483$ , matching the output of *run\_simulation.m*. Site A is selected.

## Appendix B: Safety Monitor Pseudocode

```
function SafetyMonitor(cmd, state, vehicle):
# vehicle: mass m, gear stiffness k, stroke x_max, η
Vz_max = η * x_max * sqrt(k / m)           # Section 6.1
E_max   = 0.5 * m * Vz_max**2
φ_max   = 15.0 * π / 180                    # rad
e_max   = 5.0                              # m

cmd.Vz = clip(cmd.Vz, -Vz_max, +Vz_max)
cmd.φ  = clip(cmd.φ, -φ_max, +φ_max)

if state.e_xy > e_max:
    raise 'LATERAL_EXCURSION' -> request re-score
if 0.5 * m * state.Vz**2 > E_max:
    raise 'OVER_ENERGY' -> request re-score
if state.E_required > state.E_available:
    raise 'ENERGY_INFEASIBLE' -> request re-score

log(cmd, any_raised_events)                # UK SORA / CAP 722A audit trail
return cmd
```

## Appendix C: Simulation Code and Reproducibility

The MATLAB script *run\_simulation.m* implements the full guidance pipeline used to produce Table 5 and Figures 5 to 7. It is organised in five sections: (1) vehicle and site parameters, (2) MCDM site selection using the WSM of Chapter 4, (3) Monte Carlo loop over 200 samples for both guidance configurations, (4) result aggregation and CSV export, and (5) plot generation. The random-number generator is seeded at 42 for reproducibility.

```
% Core safety monitor (from run_simulation.m, Section 3)
viol = 0; % envelope violation counter
Vz_max_nom = P.x_max * sqrt(P.k / P.m); % 3.0 m/s nominal
Vz_max = P.eta * Vz_max_nom; % 2.1 m/s at  $\eta = 0.7$ 

% Baseline guidance: aggressive proportional descent, no clamp
Vz_cmd = -0.15 * (S.z - 0);

% Constraint-based guidance: same law with safety-monitor clamp
Vz_cmd = -P.kz * (S.z - 0);
Vz_cmd = max(min(Vz_cmd, Vz_max), -Vz_max);

% Envelope violation counter (both configs, for fair comparison)
if abs(Vz_cmd) > Vz_max + 1e-6
    viol = viol + 1;
end

% Cesetti-equivalent benchmark: surface-only weighting
P.w = [0, 0, 0, 1, 0, 0]; % [wD wR wO wS wW wE]
```